

The Power of RAG

Retrieval-Augmented Generation in Data Analytics & Policy

Slide 1: The Core Problem

The Limitations of Context-Free AI

When enterprise data analysts use standard Generative AI models without providing specific documents, they expose themselves to severe risks:

Hallucination Risk

The AI attempts to guess highly specific numbers (like 2026 G7 climate budgets) based on outdated pre-training data, leading to factual errors.

Lack of Specificity

Answers become generic and broad, failing to provide the strict econometric policy constraints required for professional modeling.

Slide 2: The Solution

What is RAG? (Retrieval-Augmented Generation)

RAG grounds the AI by connecting it to private, external databases or uploaded documents before it generates an answer.

User Query → Retrieve Internal Policy Document → AI Processes Context → Grounded, Accurate Response

Result: The AI acts as an advanced reasoning engine applied *strictly* to your verified data, rather than a general search engine.

Baseline vs. Document-Grounded Responses

A side-by-side evaluation of asking about "2026 G7 Climate Finance Policies":

Evaluation Criteria	Baseline (No Document)	RAG Approach (With Document)
Accuracy	Guessed, generalized statements.	100% factually correct based on context.
Specificity	Vague mentions of "green funds".	Explicit: "40% allocation, \$450B cap, Article 4".
Grounding	None (relying on model weights).	Directly quotes and cites the uploaded PDF.
Hallucination Risk	Extremely High.	Virtually Eliminated.
Trustworthiness	Unusable for enterprise decisions.	Highly trusted for econometric modeling.

Why RAG is Mandatory for Enterprise Use

The "Golden Rule" for Data Analysts

In real-world business and data science workflows, **never rely on an AI's inherent memory for facts**. AI should be used for its *reasoning capabilities*, while your uploaded documents (via RAG) must provide the *factual knowledge base*.

- Ensures legal and policy compliance.
- Forces the AI to say "I don't know" if the fact isn't in the document.
- Transforms GenAI from a risky novelty into a reliable enterprise tool.